

Ideas Register — Consolidated, v1 to v25

One file for every parked idea, with its current status

This file replaces `ideas_v1` through `ideas_v18`. Versions 1–11 were cumulative (each carried the previous forward) and 12–18 were separate additions, so the same idea often appeared in several files at different stages. Here each idea appears once, with where it came from, where it is destined, and where it stands now.

How status was determined. Not from memory of the discussions. Each status was checked by searching the current paper, `Pattern.Arithmetic.v15.24.tex` (v15_24 differs from v15_22 only in the atom-structure passages, so earlier checks stand), and — for items destined elsewhere — the `Assembly.v1.tex` and `Inference.v1.tex` copies on hand (which may be older than your working copies; if so, the “not filed” marks below should be rechecked against yours). Entries from v19 onward are appended here; v19 to v25 are the first seven.

Status vocabulary. **CLOSED** — resolved and merged into the paper. **CLOSED (redirected)** — determined not to be the paper’s question; destination named. **RESOLVED, NOT MERGED** — fully worked out but waiting on a prerequisite. **DECIDED** — a ruling recorded; nothing to merge. **PARTLY MERGED** — some of it is in the paper. **OPEN** — still unresolved or unfiled. **PARKED** — verified but with no current home.

1 The gaps of ideas v1, as they now stand

Honest summary first, since this is what prompted the consolidation:

Gap	Status
Gap 1 — Mix across differing directions	CLOSED in v15_21: refused, on the <code>chladni</code> evidence, with Bundle and Collapse remaining available. One small residual (R26 below).
Gap 2 — Location as a sort	RESOLVED, NOT CLOSED. Fully worked out in v17 (meta vector; both encodings verified). It cannot honestly be marked closed in the paper yet: it depends on the meta-vector taxonomy being imported into Paper 1 first, and that taxonomy is not in the paper at all (zero occurrences). The paper’s own “Open: location as a sort” paragraph is therefore still correct as written.
Gap 3 — Reducing a path by merging close actors	CLOSED for Paper 1 in v15_22 (narrowed to the algebraic guarantee); the technique question is redirected to Inference and is not yet filed there.

The two remaining items from v1 are both Inference-destined and neither is filed: the candlestick summary (verified, mechanism works) and the “Inference Mandelbrot” naming flag.

The single most useful next step is importing the meta-vector taxonomy into Paper 1 (R15). It is the prerequisite for closing Gap 2, and it gives a home to location, the two time registers, and V_{derived} at once.

2 Register

ID	Item	Origin	Destination	Status
<i>Closed</i>				
R1	Mix across differing directions (Gap 1)	v1, v3–v4, v7	Paper 1	CLOSED — refused; v15.21
R2	Terminology realignment: Bundle/Remove, Collapse/UnCollapse, Mix/UnMix	v8–v10	Paper 1	CLOSED — merged; v15.20
R3	Symbols $\uplus, \setminus, \oplus, \ominus, \boxplus, \boxminus$	v9–v10	Paper 1	CLOSED — merged ($\ominus$ superseded)
R4	Collapse non-idempotent; $N =$ list length; $A^{(N)}$; UnCollapse algebra	v10	Paper 1	CLOSED — merged
R5	Distinct count D against total N	v13	Paper 1	CLOSED — merged; v15.20
R6	Reducing a path (Gap 3): algebraic guarantee only	v1, v6, v18	Paper 1	CLOSED (redirected) — v15.22
R7	τ_{weather} conflation	v4	—	CLOSED — correction recorded
R8	Same-sort region merge vs cross-sort	v6	Assembly; Paper 1	CLOSED — same-sort solved in Assembly; cross-sort is R1
R9	Internal link, terminology, cohesion audit	v12	Paper 1	CLOSED — fixes in v15.23, carried forward
R10	Stale status lines and symbols in v1–v10	v11	ideas	CLOSED — superseded by this file
R11	Inertia in Assembly via a mass-like weighting	v13	Paper 2	CLOSED — tested; fails
<i>Decided</i>				
R12	Physics names for atom metrics	v13	Paper 1	DECIDED — keep $\mathcal{E}$; N/D earned; “density” stays retired; S/N parked
R13	“uncollapse” (v5) collides with UnCollapse	v11	ideas	DECIDED — call it Mix-provenance link
<i>Resolved but not merged, and its prerequisite</i>				
R14	Gap 2: location and time as meta vectors; registers; age derived	v1, v17	Paper 1	RESOLVED, NOT MERGED — blocked on R15
R15	Meta-vector taxonomy; V_{derived}	v2, v23	Paper 1	OPEN — specified in v23; settle R45 before importing; prerequisite for R14
<i>Open — meta vectors</i>				
R16	Link vectors	v2	Paper 2	OPEN — not filed
R17	Meta-vector grouping (promote a shared one)	v2	Inference	OPEN — not filed

ID	Item	Origin	Destination	Status
R18	Identity handle / Mix-provenance link	v5, v11	Papers 1, 2, Inf.	OPEN — paper has “proposed, not adopted” only
R19	Should Mix/Collapse skip expired contributions?	v17	Paper 1; Inf.	OPEN — deliberately unanswered; v24 proposes filtering by UnCollapse
R45	Meta-vector design questions G1–G14 (v23, v24): label identity, the level rule, mutation and links, vocabulary, timestamp encoding, sentinels, granularity, comparison per kind, core multiplicity, guidance	v23, v24	Paper 1	OPEN — G1, G2, G4, G7, G8 settled in v24; G11, G13 confirmed, G12 refined, G14 revised in v25
<i>Open — Collapse</i>				
R20	Collapse across differing sorts (per-vector τ_i)	v10, v11	Paper 1	OPEN — stated in paper; broadened by R36
R21	Collapse as T ’s actor	v11	Paper 1	OPEN — “not claimed” in Axiom 7; see R40
R22	Assembly and Collapse: relation	v11	Papers 1, 2	OPEN — Axiom 6 silent; candidate in v25
<i>Open — metrics, scope, presentation</i>				
R23	Shell-indexed energy	v14	Paper 1	OPEN — undefined
R24	Antipodal ($\hat{n}$ against $-\hat{n}$) cancellation	v14	Paper 1	OPEN — special case of closed R1
R25	Separable-metric pointwise test	v15	Paper 1	PARKED — verified; value depends on R27
R26	Route-1 scoping clause (refusal is about $\boxplus$ only)	v7	Paper 1	PARTLY MERGED — one sentence not carried; closure wording “already demonstrated ... correct behavior” overstates (v22)
R27	Axiom 7 metric catalogue and costumes: cut or move?	v16	Paper 1	OPEN — undecided
R28	Two-paper split	v16	Papers 1, 2	OPEN — undecided
R29	Review’s presentational items	v16	Papers 1, 2	OPEN — not acted on
<i>Open — Inference-bound</i>				
R30	Candlestick summary	v1	Inference	OPEN — verified; not filed
R31	“Inference Mandelbrot” naming flag	v1	Inference	OPEN — flag only
R32	Technique selection for reduction (average, centroids, shape-preserving)	v18	Inference	OPEN — specified; not filed
R33	Douglas–Peucker adapted to the sphere	v18	Inference	OPEN — the one unbuilt piece
R34	Sphere-packing bound for the “stated distance”	v6	Inference	OPEN — not filed anywhere
R35	Coherence check for the candlestick	v1	Inference	OPEN — never considered

ID	Item	Origin	Destination	Status
<i>Open — atom structure (v19)</i>				
R36	Pure and mixed atoms: cores; per-vector τ_i as the general form; Mona Lisa as test case; direction-based derived quantities per sort	v19	Paper 1	OPEN — broadens R20
R37	Collapse of multi-vector contributions (not fixed at one vector each)	v19	Paper 1	CLOSED — cores of any size, kept distinct under Collapse; merged in v15.24
R38	N as number of cores; Q as number of vectors (resolves the collision with N as vector count in Axiom 7)	v19	Paper 1	CLOSED — N cores, Q vectors; merged in v15.23
R39	Collapsed atom as a counted bowl; commuting test	v19	Paper 1	OPEN — test not yet run
R40	T asymmetry: a bowl is not an actor; a Collapsed atom needs a cast	v19	Paper 1	OPEN — relates to R21
R44	Empty atom: black is $N = 1$, $Q = 0$; empty core under Collapse; zero-strength vector is not an arrow; conventions at $Q < 2$	v22	Paper 1	CLOSED — merged in v15.24; “info number” label moved to Q (confirm)
<i>Open — energy and effective counts (v20)</i>				
R41	Energy from strength, vector count and direction: $\mathcal{E} = S^2/Q + Q \text{Var}(r)$; $S^2/Q \leq \mathcal{E} \leq S$; $ V ^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$	v20	Paper 1	PARKED — identities verified, not in paper; $\mathcal{E}$ keeps “energy,” $\mathcal{V}$ keeps “volume”
R42	Effective vector count $Q_{\text{eff}} = S^2/\mathcal{E}$; overlap with σ_{atom}	v20	Paper 1	CLOSED — merged in v15.23 as a re-expression, not an eighth metric; both it and σ_{atom} kept
R43	Strength-only metrics per sort and for the whole; effective number of sorts $1/\sum w_s^2$; pooling ratio	v20	Paper 1	OPEN — needs one stated convention

3 The entries, by version

v1 — the three gaps and two Inference items

Gap 1 (Mix across differing directions): Axiom 5 defines Mix only at a shared $\hat{n}$; a provisional vector-sum recipe had been tried and withdrawn. *Status: R1, closed.* **Gap 2** (location as a sort): a third candidate was tested — disjoint dimensions for content, time, and location. Verified: two shifts through disjoint dimensions commute exactly (to full floating-point precision), and content stays recoverable by projection; shifts forced onto one shared dimension do not commute. Left open: the encoding itself. *Status: R14, resolved in v17.* **Gap 3** (reducing a path): needed a distance, a merge rule, and a drift bound. Verified that swapping the clip for the Hill form $r' = |z|/(K+|z|)$ changes the strength but not $\chi' = \arg(z)$. *Status: R6, closed (redirected).* **Candlestick summary:** open/close/high/low distinguishes two sequences of identical content in opposite order

that naive Mix cannot — Mix gave the same 47.5° for both; the candle’s open and close swapped ($20^\circ \rightarrow 160^\circ$ against $160^\circ \rightarrow 20^\circ$) while high and low stayed equal. *Status: R30, 0 occurrences in Inference_v1.* **“Inference Mandelbrot”**: an ordered-thing promotion is a different problem from Assembly’s Hill-bijection promotion of unordered camps, and needs its own name. *Status: R31.* The “considered and found to need nothing filed” list (reference atoms, precursor-witness relabeling, the $\pi = (x; w_1, \dots, w_n)$ trajectory, χ as internal phase-color) was rechecked in v11 against the current paper: all present, nothing to file. *Closed.*

v2 — meta vectors, split across three papers

Proposed: besides information vectors (capped, $r_i \leq 1$), an atom may carry meta vectors describing it rather than being part of its content: V_{derived} , link vectors, a timestamp vector. Meta vectors are not capped. Split by destination: the taxonomy and V_{derived} to Paper 1; link vectors to Assembly; the grouping mechanism (query nearby meta vectors, promote one shared value) to Inference. Verified caveat: grouping V_{derived} inherits the Hill form’s limit — both fix magnitude only, never direction. *Status: R15–R17. The paper has zero occurrences of “meta vector”; Assembly_v1 has zero of “link vector.”*

v3 — Gap 1, characterized

Argued reading: Mix takes several reports already agreed to answer one shared question and returns a verdict about it; the shared $\hat{n}$ is what makes that true. Notably, v3 already stated that in Mix’s domain “ $\hat{n}$ itself was never plural; only χ ever varied” — the very distinction that had to be rediscovered during the Gap 3 discussion (v18) before it stopped causing confusion. Four “we mix all the time” counterexamples were checked and dissolved (sugar in tea is one dial; red plus green is already same- $\hat{n}$; paint compositing is a bowl and flattening is destructive and irreversible — matching UnMix’s impossibility). The one unresolved example, a whole painting as one vector, is Gap 1 itself. The open question split in two: *embedding validity* (does closeness on a shared sphere track semantic agreement?) and the *merge rule*. *Status: superseded by R1’s closure — Mix refuses, so neither sub-question needs an answer for Mix.*

v4 — the τ_{weather} correction; Gap 1 “ready to close”

τ_{weather} had named two different problems: combining different weather concepts across directions (Gap 1), and reducing closely aligned same-sort atoms to one representative, which is Assembly’s already-verified Hill-bijection mechanism (16 camps to 13 systems). The **chladni** section was found to already contain the evidence for Gap 1 (“cross-mode coexistence is therefore $\oplus$ ”). *Status: R7 closed; the chladni evidence is what the Gap 1 closure rests on.*

v5 — identity handle

An opaque *id* meta field, assigned before Bundle or Mix and kept after, so a result can point back to the bowl that produced it (bowl {red, green} is #17; Mix gives yellow #18 with a link $\#18 \rightarrow \#17$). Refuse τ_{id} as a sort; keep *id* as a field. Navigation by a kept record, not inversion of Mix. Renamed in v11 “Mix-provenance link” because the original word “uncollapse” collides with the formal UnCollapse operator. *Status: R13 decided; R18 open — the paper mentions id_i once, as “proposed, not adopted,” in UnCollapse’s well-definedness paragraph.*

v6 — sphere-packing bound; merging a whole region

For Gap 3’s “stated distance”: the number of mutually distinguishable points at separation $\theta_{\min}$ is roughly $2/(1 - \cos(\theta_{\min}/2))$ — about 52,525 at 1° , 2,101 at 5° , 526 at 10° , 59 at 30° , 15 at 60° , 7 at 90° , 4 at 120° . Merging a whole region to one representative splits: same-sort is solved (Hill bijection) and cross-sort is Gap 1. *Status: R8 closed; the bound itself is R34, filed nowhere.*

v7 — scoping of Route 1

“Refuse outright” should be worded as a claim about $\boxplus$, not a permanent bar on any future, separately admitted mechanism — the same “undecided, not admitted” logic Axiom 0 already uses. *Status: R26, partly merged.* Checked against the Gap 1 closure: the closure is scoped to Mix and keeps $\boxplus$ and $\oplus$ available, which honours the point. What was not carried is the one explicit sentence that a separately admitted mechanism is not barred, and the closure calls the other two routes “withdrawn,” slightly stronger than v7 intended. An optional one-sentence fix.

v8, v9, v10 — terminology and verb inventory

v8 proposed the Collapse/UnCollapse/Bundle/Mix/UnMix realignment and flagged a hard symbol collision ($\ominus$ was Remove). v9 chose $\boxplus$ for Bundle and $\ominus$ for UnCollapse, and scoped ids to the one case that needs them: duplicate-valued contributions in UnCollapse. v10 revised the symbols — $\ominus$ to UnCollapse, Remove to $\setminus$ — settled Bundle’s idempotence ($A \boxplus A = A$, already proven), and showed Collapse is *not* idempotent by three convergent arguments (information loss; an algebraic contradiction with $A \text{ UnCollapse } A = 0$; the same evidence-count blindness the Hill form was built to fix). N is simply the length of the vector list; $A^{(N)}$ abbreviates N identical copies; UnCollapse sends $N \rightarrow N - 1$. *Status: R2–R4 closed, all merged. MixMerge was later reverted to Mix at your request; $\ominus$ appears zero times in the paper. The per-vector τ_i question that v10 left open is R20.*

v11 — review of v10 against the paper

Status lines of v8–v10 corrected (the realignment was merged); stale $\oplus/\ominus$ readings in earlier text fixed, including one sentence that had become *false* (“Gather still satisfies $A \oplus A = A$ ” reads wrongly once $\oplus$ means Collapse); the “uncollapse” collision resolved by renaming. Three gaps opened by giving Collapse axiom-level standing: R20 (across sorts), R21 (as an actor for T), R22 (no stated relation to Assembly — Axiom 6 mentions Collapse zero times, so this one is silence rather than a stated omission). One new route recorded: Collapse losslessly first, then define a reading — the seed of what v18 developed. *Status: R10, R13 closed or decided; R20–R22 open.*

v12 — internal audit

96 labels and 198 reference call-sites checked, each against what its sentence claims; zero mismatches. Two stragglers fixed (an Abstract “a Collapse” noun phrase; a Forest-costume “gatherers”). Four cohesion fixes where the paper’s own summaries omitted Collapse, and the UnMix proof section now names UnMix. No new conceptual gap. *Status: R9 closed; fixes are in the current paper.*

v13 — naming the atom metrics

Seven metrics, quoted precisely. Earlier drafts had called $\mathcal{V}_{\text{atom}}$ “mass,” γ_{atom} “density,” and S_{atom} “atom energy,” and retired all three; only $\mathcal{E}_{\text{atom}} = \sum r_i^2$ kept “energy,” because it is tied by an

exact identity to $|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)$ (variance-of-a-sum structure, verified to six decimals). The rule: a physics name is kept only if the quantity participates in something provable. $D = |A_1 \uplus \dots \uplus A_n|$ passes (forced by $\uplus$'s idempotence), so N and D keep the mass-number/atomic-number analogy; “density” stays retired; S_{atom}/N is parked rather than named. Inertia in Assembly is closed: a mass-like weighting was already tested in the motion equation — a ninefold strength gap gives a 729-fold weight and a non-monotonic trajectory. *Added in conversation after v15, not previously filed:* asked whether either of the two metrics survives, the answer was that energy already does ($\mathcal{E}_{\text{atom}}$, earned) while no candidate for density does — S_{atom} already has a name, N is a count rather than a ratio, and the one ratio with the right shape, γ_{atom} , is the one already rejected. *Status: R5 closed; R11 closed; R12 decided.*

v14 — three rough sketches

Density restated: nothing new, closed by reference to v13. *Energy as shell distribution:* partly already true of $\mathcal{E}_{\text{atom}}$ — for fixed total strength $S = 2.0$, four vectors give $\mathcal{E} = 1.0$ spread evenly and 1.81 concentrated toward the extremes (and σ_{atom} goes $0 \rightarrow 0.333$) — but σ_{atom} is blind to *which side* the concentration is on, and no shell-indexed quantity exists. *Matter/antimatter:* antipodal cancellation ($\hat{n}$ against $-\hat{n}$), a second kind of opposition beside the existing same-direction opposite-phase cancellation. With Gap 1 now closed by refusal, this would be a deliberate special-case exception, parallel to the existing one; what the rule would produce, and whether it needs equal strength, is undecided. *Status: R23, R24 open.*

v15 — when two metrics constrain each other

For separable metrics $M_1 = \sum f(r_i)$ and $M_2 = \sum g(r_i)$: if $f \leq g$ everywhere on $[0, 1]$ then $M_1 \leq M_2$ for every atom; if the curves cross there is no bound. Verified against both cases (it forces $\mathcal{E} \leq S$ because $r^2 \leq r$; it forces nothing for N against a shell-weighted energy). Where a bound holds, the gap is itself a metric ($r - r^2$, largest at $r = 0.5$). Possible uses (a non-redundant reporting set, a two-axis placement, a debugging check, a post-hoc label for Assembly's camps) are speculative. *Status: R25, parked; it matters mainly if R27 keeps the metrics.*

v16 — external review and the two-paper split

Every checkable claim in the review was verified: the worked numbers (e.g. $r' \approx 0.721$ at 73.9° ; freeze at 628 and 1261 steps), that all 22 occurrences of the six Axiom 7 metrics sit inside Axiom 7 and none in Bundle/Mix/T/offset, that “white” is used in two senses (the paper already says so in Axiom 3), and that the “checked directly” precision claims have no supplement. The proposal: Paper 1 is the typed algebra, Paper 2 is assembly and the enclosure, the Axiom 7 catalogue and the costumes go in neither; Paper 1 states Assembly as a signature only and ends by naming the seat it refuses to supply; cut by question, not by length. *Not itemized when v16 was filed, but in the review and still pending (R29):* state the clip, the Mix refusal, and the assembly energy as stipulated choices in one place rather than narrated as discoveries; print the enclosure gradient; provide a supplement for the “checked directly” claims; and stop calling the enclosure a “law” in the same voice as $\text{SO}(3)$ while it still needs δ , η , the census cut, and $\varepsilon_{\text{allow}}$. *Status: R27, R28, R29 open and undecided.*

v17 — Gap 2 resolved

Meta vector, not sort, tested against the cap: $r \leq 1$ encodes strength of content, while location and time are coordinates. This is the line the paper already draws for Assembly’s seat. Location is one vector via standard (θ, ϕ) from latitude and longitude (London $\rightarrow (0.6225, -0.0011, 0.7826)$, Tokyo $\rightarrow (-0.6194, 0.5252, 0.5835)$, Sydney $\rightarrow (-0.7273, 0.3999, -0.5577)$). Time on (θ, ϕ) fails: at hour 0 every minute lands on $(0, 0, 1)$, and 11:59 and 0:00 land about 2 apart (antipodal). The fix: minute on ϕ , hour on χ , θ fixed at the equator (6° and under 1° apart for the same pair). Timestamp is a register of length N , kept in step with UnCollapse; age is derived (now – timestamp), never stored; expiry is a second register because it cannot be derived; timestamps help but do not guarantee disambiguation the way an *id* would. **Blocking:** cannot be merged until R15 is. *Status: R14, R19.*

v18 — Gap 3 closed for Paper 1

Corrected an earlier confusion (χ is phase at a shared direction, not $\hat{n}$; Gap 1’s refusal did not apply). The real issue is order-blindness: merging three actors in either order gives the same result. A path’s commitment to order comes from outside the algebra — Bundle discards order, and choosing T over a bowl is the declaration that order matters. So averaging 10,000 temperature readings is bowl-plus-Mix and loses nothing. Three techniques answer three different questions: averaging cannot tell a spike from a flat line; centroids give identical results for a sequence and its reverse (clustering is blind to position); shape-preserving reduction keeps real points at their positions — Douglas–Peucker, whose tolerance is the drift bound Gap 3 asked for. Paper 1 was narrowed to the non-consumption guarantee. *Status: R6 closed (redirected); R32, R33 open for Inference.*

v19 — pure and mixed atoms, cores, and Collapse against a bowl

Prompted by a hypothetical: could the Mona Lisa be held in one atom? “One atom” can mean three things. A *name*: a single seated point, like “storm,” which the paper calls a doorbell — it carries none of the content. A *bowl*: already legal for mixed sorts, since Bundle is sort-blind. A *Collapsed atom* holding colors, words, and strokes, which needs mixed sorts inside one atom. This is the example v3 left unresolved (“a whole painting as a single vector” was Gap 1 restated). With Gap 1 closed, a single *vector* is refused, while one atom of many vectors stays available. What stands in the way of the third meaning: mixed sorts (R20, widened to R36); where each contribution sits on the canvas — a location per contribution, the same kind of register as v17’s but not yet specified per contribution; derived quantities across sorts; and relations, since the paper “does not mint the relations” and, by its own design rule, which units were co-present and within what region are facts of a level, not coordinates of an atom.

Rulings given in the discussion, none yet in the paper.

- *Collapse juxtaposes.* Positions do not change, nothing merges, no new type is minted. Relations within a sort are as before; across sorts they exist only under a stated rule (Axiom 0’s “undecided, not admitted,” applied to Collapse). So derived quantities that use direction are computed per sort, and cross-sort ones are undefined until a rule exists. (Refined in v20: strength-only quantities may also be given for the whole atom.)
- *Cores.* A core is a unit inside an atom, usually one τ (a pure type) but free to hold several — words and numbers together — as typical, not as a rule. The general form is per-vector τ_i , with “core” a grouping. “Core” appears in the paper only in the “Core Contributions” heading and in “no core, no halo” in the enclosure, so it needs one defined sense.
- *Collapse is not fixed at one vector per contribution.* This settles the question the paper was silent on: whether collapsing already-multi-vector atoms flattens or nests. The Collapse text still says “one vector per contribution.”
- *N is the number of cores.* This keeps UnCollapse’s $N \rightarrow N - 1$, keeps D a bowl size (distinct cores), and keeps the v17 registers at length N , so those registers are per core, not per vector. The cost is a word collision: N is currently the vector count in Axiom 7, in $\gamma_{\text{atom}} = \mathcal{V}_{\text{atom}}/N$ and in σ_{atom} ’s $N^2/(N - 1)$. *Decided:* N counts cores and Q counts vectors. All seven Axiom 7 metrics run over vectors, so they take Q ; the D/N paragraph and the UnCollapse text keep N and change “vectors” to “cores.” To settle at the merge: the sentence that defines N as the atom’s count of informational vectors, and where the label “info number” goes.
- *A Collapsed atom matches a counted bowl.* They are identical exactly when nothing repeats ($D = N$), and $N - D$ is the gap. The paper’s bowl is a set, and its own counting equation shows the difference: $A \uplus A \uplus A \uplus A \setminus A = \emptyset$, while $A \oplus A \oplus A \oplus A \ominus A = A^{(3)}$ (by the UnCollapse rule). Operations should work on both; recorded as a commuting test — a pack/unpack map between the two, with every operation required to agree across it. Sum-style quantities (S , $\mathcal{E}$, V_{derived}) agree only on a counted bowl. T is asymmetric today: it “does not act on the set R as a set,” and a Collapsed atom can act only once a cast is named.
- *Two distinct ways of arranging and viewing vectors.* They differ in multiplicity (set against counted), in handle (a Collapsed atom is one unit that can sit in a bowl or at a lexicon pin; a bowl is a collection), and in sequence (neither stores order; the path carries it, and a bowl is a good surface for displaying one).

Status: R36–R40; R20 and R21 cross-referenced.

v20 — energy from strength, counts, and direction; effective counts

Started from a recollection that energy is $\sum r^3$. The paper’s energy is $\mathcal{E} = \sum r^2$; $\sum r^3$ is the atom’s volume, formerly “mass.” Nothing needs renaming, and $\sum r^3$ keeps a geometric reading (strength as a radius, as in Assembly’s softening length).

Energy from strength, vector count Q , and direction already exists in $\mathcal{E}$. Three exact relations, verified on 20,000 random atoms (errors near 10^{-14} , no violations), with $S = \sum r_i$:

- $\mathcal{E} = S^2/Q + Q \text{Var}(r)$: an evenly spread floor, set by S and Q alone, plus concentration.
- $S^2/Q \leq \mathcal{E} \leq S$.

- $|V_{\text{derived}}|^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$: the resultant’s energy, blending the incoherent baseline with the fully aligned value, weighted by the paper’s coherence α .

Which works better: the floor S^2/Q cannot tell an even atom from a concentrated one (both give 1.000 at $S = 2$, $Q = 4$, where $\mathcal{E}$ gives 1.000 against 1.810). $\mathcal{V} = \sum r^3$ separates them even more sharply (0.500 against 1.715) but has no identity tying it to anything beyond the radius reading. So $\mathcal{E}$ is the energy and $\mathcal{V}$ stays volume. Cores N enter through the same variance-of-a-sum algebra applied per core (plain algebra, not run separately) and through the ratios Q/N and S/N . The mean strength S/Q , the parked candidate from v13 (then written S/N), is the centre of the variance decomposition and keeps a plain name.

Effective vector count $Q_{\text{eff}} = S^2/\mathcal{E}$ (the participation ratio in physics, the effective sample size in statistics).

- Q counts vectors and Q_{eff} counts those that matter: 100 vectors at 0.05 give 100; one vector at 0.9 plus 99 at 0.001 give 1.23.
- The largest vector carries at least $1/Q_{\text{eff}}$ of the strength (50,000 random atoms, no violations).
- Scale-free: halving every r leaves it unchanged (2.2099 both).
- With direction: $|V|^2/S^2 = \alpha + (1 - \alpha)/Q_{\text{eff}}$ (error near 7×10^{-16}), so two dimensionless numbers fix the resultant’s size.
- Limits: it re-expresses S and $\mathcal{E}$, adding no information, and ignores direction.
- Overlap: $\sigma_{\text{atom}} = (Q/Q_{\text{eff},\mathcal{V}} - 1)/(Q - 1)$, where $Q_{\text{eff},\mathcal{V}}$ is the effective count of the *cubed* shares (error near 10^{-15}). So σ_{atom} is already this family, and Q_{eff} is its plain-strength sibling, the one tied by identity to $\mathcal{E}$ and $|V|^2$. Whether to keep both, one, or neither is undecided.

Per sort and for the whole. The per-sort rule belongs to quantities that use direction (V_{derived} , α , $|V|^2$), since sums and dot products of directions across sorts mean nothing. Strength-only quantities (S , $\mathcal{E}$, $\mathcal{V}$, Q_{eff} , the σ ’s) can be given per sort and for the whole, on one stated convention: each sort’s r is normalized to its own declared saturation (the paper’s $r = a/a_{\text{max}}$), and whole-atom figures treat fraction of own saturation as a common scale. With w_s the share of total strength held by sort s :

- Pooling ratio $\sum_s Q_{\text{eff},s}/Q_{\text{eff,whole}} \geq 1$ (Cauchy–Schwarz; no violations in 30,000 atoms), with equality exactly when every sort has the same strength-weighted mean strength $\mathcal{E}_s/S_s$. It measures imbalance between sorts: two vectors at 0.9 against ten at 0.1 give 2.63.
- Exact decomposition: $1/Q_{\text{eff,whole}} = \sum_s w_s^2/Q_{\text{eff},s}$ (error near 2×10^{-16}). In the symmetric case it multiplies: three sorts of four vectors at 0.5 give $12 = 3 \times 4$.
- Effective number of sorts $1/\sum_s w_s^2$ answers “is this atom really mixed?” more readably than the ratio: for 50 color vectors at 0.5, 3 word vectors at 0.5, and 10 stroke vectors at 0.3, Q_{eff} is 50, 3, 10 per sort, 61.5 for the whole, and the effective number of sorts is 1.37. Three sorts are present but the atom reads as about 1.4, because color carries 85% of the strength.

Status: R41–R43; the v19 per-sort line is refined above.

v21 — merging N , Q , and Q_{eff} into the paper

Merged into `Pattern_Arithmetic_v15_23.tex`. Before any edit, all 62 standalone uses of N were audited and classified, since a blind replacement would also have hit N_{dim} and N_{vox} .

What went in. “Core” is defined once, at Axiom 7’s opening: N counts cores (the atom’s info number) and Q counts informational vectors, with $Q \geq N$. Every formula that runs over vectors takes Q : the V_{derived} sum, S , $\mathcal{V}$, γ , $\mathcal{E}$, the coherence and concentration discussions, σ_{atom} , σ_{shell} , the higher-dimensional actor condition $Q \geq N_{\text{dim}} - 2$, and the closing teaching-object sentence. The Collapse, UnCollapse and D/N text keeps N and now says “cores,” and UnCollapse states that Q falls by the removed core’s vector count. Q_{eff} is defined at the end of Axiom 7 as a re-expression of S and $\mathcal{E}$, with its range, its scale-freeness, the largest-vector bound, the count-versus-evenness distinction, and its exact relation to σ_{atom} .

Found along the way. The audit turned up more than the rename:

- N was also the clock modulus in the numeric-sort definition, while the later clock section already used $M = 24$. The general form now uses M .
- “Core” was already in the paper twice: the heading “Core Contributions” (now “Principal Contributions”) and “no core, no halo” in the enclosure (now “no dense center, no halo”).
- A stray noun use, “the actor’s Collapse,” survived every earlier terminology sweep because the search pattern missed possessives. It now reads “Collapsed atom.”
- σ_{atom} is exactly the normalized Herfindahl index of the cubed shares, $(QH - 1)/(Q - 1)$, and the paper had already rejected the unnormalized index for reading an even atom as $1/Q$. This is why Q_{eff} is described as a *count*, with Q_{eff}/Q as the size-free form. It sharpens v20, which called the two overlapping: they are the same construction on different shares.

Deliberately not merged. Mixed sorts, per-vector τ_i (R36; R20 stays open and the Collapse section still requires a shared sort); collapsing atoms that already hold several cores, now stated in the paper as not defined; the per-core registers (R14, R15); and the other R41 identities.

Checks. The 18 numbered equations before Process Principles are identical to v15_22’s after mapping Q back to N , so the math changed only by the symbol. Zero errors, zero undefined references, and the same five overfull warnings as before. One choice to confirm: the label “info number” was attached to N . *Status: R38 and R42 closed; R37 partly merged.*

v22 — cores of any size, the empty atom, and a review of v15_23

Merged into `Pattern_Arithmetic_v15_24.tex`. The trigger was an external critique of v15_23, checked claim by claim against the text before anything changed. (The v21 entry above reached this file only now; it describes v15_23, which this entry supersedes where they differ.)

The critique, as it came out.

- *Confirmed.* No example separated N from Q and, stronger than the critique said, no construction in the paper could produce $Q > N$: the lexicon cast seats single points, and collapsing multi-vector atoms was “not defined here.” The UnCollapse well-definedness paragraph and Table 2’s footnote still spoke of vectors where cores were meant. D was undefined for multi-vector cores: with $A = \{0.8, 0.4\}$ and $B = \{0.6\}$, $\text{Collapse}(A, B, A)$ has $D = 2$ read as a bowl and $D = 3$ read as distinct vectors.
- *Half right.* The nucleon analogy: nucleon = core was the intended mapping, but the sentence did not say so.

- *Not valid.* That clipping is narrated as a finding: the paper has a heading “How the bound is imposed is a choice.”
- *Valid, and my wording.* The Mix-refusal closure calls refusal “already demonstrated, in a concrete domain, to be the correct behavior,” which overstates what `chladni` shows. Recorded against R26.
- *Missed by the critique, found in the check.* The five-fold cone sentence was ambiguous: “the same half-angle” reproduces the printed 7.14 and 0.988 only at half-angle 5.7° , while the pair-centred reading gives 7.203 and 0.997. Also, n and N both named the operand count, and Q_{eff} was undefined at $S = 0$.

Process note. The v21 audit searched for N and not for the word “vector,” so passages that never contain N were invisible to it. This round searched for both.

Rulings applied. $Q \geq N$ is the usual case but not a rule, and $N = Q = 1$ is the teaching atom, one state among several. An atom may hold no vector at all: the empty atom is black, one seat and no arrow, $N = 1$, $Q = 0$, which is Axiom 2’s own wording. Three consequences were confirmed: black is a counted contribution under Collapse ($A \oplus 0$ adds a core and no vectors, so $0 \oplus 0 \neq 0$); Collapse keeps cores distinct ($N = \sum_i N_i$, $Q = \sum_i Q_i$), which answers flatten against nest; and UnCollapse at $N = 1$ leaves black, the one exception to $N \rightarrow N - 1$.

What changed in the paper. Axiom 7’s opening restated around cores; a sentence in Axiom 2; the Collapse definition; non-idempotence stated for every A , black included, with $A^{(k)}$ replacing $A^{(N)}$; a paragraph on silent contributions; D defined by core identity (two cores are one value exactly when $\uplus$ would seat them together, and empty cores coincide); a handle per core; the analogy wording; Table 2’s footnote; two summary sentences; the cone’s half-angle; Q_{eff} stated for $S > 0$; an “empty atom” paragraph of conventions; and a worked count, $N = 3$, $Q = 5$, $D = 2$, with UnCollapse leaving $N = 2$, $Q = 3$ and a silent contribution giving $N = 4$, $D = 3$, $Q = 5$. The count is given in counts only: a first draft quoted strengths and put the seven metrics outside Axiom 7, undoing the isolation the review verified, so it was reverted.

Two consequences applied that were not discussed, to confirm. First, the label “info number” now goes with Q , since Q counts the information and adding an empty core does not change it; v21 had attached it to N . Second, informational vectors are positive, $0 < r_i \leq 1$: a vector of strength zero is black and not an arrow, so it is the empty core and does not count in Q . This follows from Axiom 2 and avoids a zero-strength vector counting in Q while black does not.

Checks. All 49 numbered equations are identical to v15_23’s; zero errors and undefined references; the same five overfull warnings as before; the seven metrics and Q_{eff} occur zero times outside Axiom 7. *Status: R37 closed; R44 closed; R26 extended.*

v23 — meta vectors as specified, and what to settle before importing them

Specification as given, parked here and not yet in the paper:

- Meta vectors are labels on coaches; information vectors are the content in the coaches. Meta vectors are not capped at 1.
- Well-known ones: V_{derived} (the sum of the vectors), $V_{\text{timestamp}}$, V_{expiry} (0 meaning no expiry), $V_{\text{lastmodify}}$, V_{location} , and V_{link} (forward, backward, bidirectional).
- They give information atoms kinds that help group them for specific tasks; the applications named are typing and sorting.

- They can build data structures in Assembly (sequences, graphs, layered clusters) and in Inference.
- They are part of information atoms. Some change with content and some do not.

Review of that specification against the paper’s own text, with the demonstrations run. Nothing in the paper changed.

Conflicts with text already in the paper.

- *G1, label identity.* $\uplus$, D and UnCollapse compare content. If a timestamp joins an atom’s identity, every atom is unique by birth time: three contributions A , B , A give $D = 2$ by content and $D = 3$ with timestamps, so $N - D$ is always 0 and $A \uplus A \neq A$. Equality must be content-only, and a rule is needed for which labels survive when $\uplus$ merges two atoms of equal content.
- *G2, the level rule.* §3.1 says “a unit carries no representation of what it constitutes. A seated word does not know the poem,” and that which units were co-present, over what interval, and within what region are “constituting facts of a level, not coordinates of an atom,” which is “why the atom stays at four coordinates and one sort tag.” Time, place and links sit against this. A candidate resolution: labels ride beside the four coordinates, as V_{derived} already does, and a link may point to sources or peers but not up to what the atom constitutes, since a word’s forward link to its poem would break the rule. The sentence needs rewording either way.
- *G3, the derived-vector limit.* Axiom 7 says “at most one further, derived vector may sit alongside them.” A family of labels contradicts it. Reword to at most one derived-from-content vector (one per sort once R36 lands), with recorded labels separate, all excluded from N and Q , and the seven metrics and V_{derived} ranging over informational vectors only. $V_{\text{derived}} = \sum r_i \hat{n}_i$ with χ excluded, which “the sum of the vectors” abbreviates.
- *G4, mutation.* The paper has no in-place change: its verbs act “alongside, not in place of,” and Collapse produces an additional atom. $V_{\text{lastmodify}}$ and forward or bidirectional links assume an atom that persists through change, which is the identity handle (R18, not adopted). With immutable atoms only backward links are free, since a newer atom can point to older sources, which is the provenance reasoning the Collapse section already makes; $V_{\text{lastmodify}}$ then equals the newer atom’s $V_{\text{timestamp}}$. One world must be chosen.
- *G5, vocabulary.* “Typed” is in the paper’s title, and type/typed occur 21 times and sort 88 times. Meta vectors giving atoms “types,” and “sorting” them, collide with τ . Use another word for the kind (label or class) and “order” or “rank” for the verb.

Gotchas in the logic.

- *G6, timestamp encoding.* v17’s clock-face vector gives the identical vector for now, one day ago, and one week ago (checked), so age and expiry cannot be read from it. They need an absolute, non-cyclic time. This answers the earlier “one component or two”: an absolute scalar for age and expiry, and optionally the clock-face vector for grouping by time of day.
- *G7, sentinels.* “Expiry 0 means none” reads as already expired under $\text{now} < \text{expiry}$ (checked), and a location of $(0, 0)$ is a real place. The paper’s own idiom is that black is an empty seat, so “no expiry” is the absence of V_{expiry} , and absence is not zero.

- *G8, granularity and propagation.* Coach may mean atom or core. Timestamp, expiry and link are per core (the R38 registers), while V_{derived} is per atom, and per sort once R36 lands. A Collapse result is a new atom whose own birth differs from its cores' births, so there are two time notions. Each verb also needs a stated rule for the labels of its result; the proposal is that a result is new with its own birth, core labels travel with their cores under Collapse, and links record provenance.
- *G9, grouping needs a comparison per kind.* Sphere distance for location, cyclic distance for the clock, equality for identifiers, graph distance for links: no common metric, so each kind needs a stated rule, the analogue of Axiom 0 for labels. Links are identifiers and not coordinates, and calling labels vectors invites arithmetic; no verb acts on a label, as none acts on the seat $\vec{P}$.
- *G10, two sources of structure.* Links declare structure, while Assembly's energy produces camps from geometry and a seat; keep them distinct. Link use belongs to Paper 2 (R16) and the taxonomy to Paper 1. Links held in labels can also carry order past $\uplus$, so state that $\uplus$ does not read labels, which keeps "order comes from outside" true of what $\uplus$ reads.

Status: R15 specified; R45 open.

v24 — rulings on the meta-vector questions

Rulings given:

- Drop $V_{\text{lastmodify}}$. Atoms are immutable and change only through Collapse and UnCollapse.
- Meta vectors are not part of D .
- Meta vectors link to other things mostly by criteria on other meta vectors and never by content. Words are never linked to a poem: the words do not know the poem, the path knows it.
- The absence of V_{expiry} means no expiry.
- Coach means atom, very loosely. Information atoms and meta vectors share the same cores, a core being a single point whose weight varies with the N count; V_{derived} has a magnitude and a direction that help in higher dimensions.
- Meta vectors help guide information atoms in higher dimensions, a useful attribute in Assembly and Inference.

What each ruling settles, checked against the paper.

- *G1, label identity.* Equality is content-only: two cores coincide when they hold the same vectors, and labels do not enter D , so they do not enter $\uplus$'s equality or UnCollapse's value-exactness either. What $\uplus$ does with the labels of merged atoms remains (G14).
- *G2, the level rule.* It stands unchanged. Links run from label to label, and a link never carries membership in a larger whole, which is the paper's own rule that a seated word does not know the poem.

- *G4, mutation.* Immutable atoms make backward links the free ones, and $V_{\text{lastmodify}}$ is dropped. One wording consequence: the paper says UnCollapse “removes one core from a Collapsed atom,” which reads as mutation. It returns a new atom without that core and leaves the original unchanged, a one-line fix at import.
- *G7, sentinels.* Absence is not zero, and the rule extends to every label. Validity is then: valid exactly when V_{expiry} is absent or now $< V_{\text{expiry}}$.
- *G8, granularity.* Shared cores make labels per core, matching the R38 registers. V_{derived} exists per core and per atom, and the atom’s is exactly the sum of its cores’, since $\sum_i r_i \hat{n}_i$ regroups by core.

New points the rulings raise.

- *G11, a core as a point with weight.* A multi-vector core spans several directions, so the single point is its V_{derived} , not its content. The weight at a point is the multiplicity $m(c)$, the number of equal cores there, with $N = \sum_c m(c)$, D the number of distinct cores, and $N - D = \sum_c (m(c) - 1)$. Call it multiplicity, not density, which R12 retired, and keep it a property of the atom’s list and not of the core: a core does not know its siblings.
- *G12, V_{derived} as a guide.* The paper already says operators needing one pair read the derived sum as a weight or a direction, and that beyond three dimensions it cannot be T ’s actor, which needs $N_{\text{dim}} - 2$ orthonormal arrows. So guidance means read-only roles: a representative weight or direction, a seat, grouping. Its reliability is atom coherence, $|V_{\text{derived}}|/S_{\text{atom}}$, which is 0 for an equal antipodal pair although $S > 0$, leaving the direction undefined. The verified third candidate of v1, disjoint dimensions, which commute exactly and keep content recoverable, is the available mechanism for labels to occupy their own dimensions without disturbing content.
- *G13, links.* Links based on label criteria are derived at read time. Only backward provenance, identity-based since the identifier is itself a label (R18), is stored at birth. Forward and bidirectional are views from a reverse index, and the three V_{link} names become view directions.
- *G14, $\uplus$ and labels.* $\uplus$ discards order and count, so it naturally discards per-copy labels too, while Collapse keeps every copy with its labels, which is the per-core register. This is the proposed default and is unconfirmed. It also resolves R19 by proposal: filtering by expiry is UnCollapse of the expired cores, a selection by label criterion, so Mix and Collapse need not know about expiry.

The paper already holds two meta vectors without the name: V_{derived} and the seat $\vec{P}$, which is uncapped, sits outside the four-tuple, and enters $d_{AB} = |\vec{P}_B - \vec{P}_A|$. Importing the taxonomy therefore mostly names what is there, and V_{location} may turn out to be the seat.

Still open before import: the wording of the derived-vector limit (G3), the word for the kind (G5), the timestamp as an absolute scalar with an optional clock-face vector (G6, unconfirmed), the comparison rule per kind (G9), and the two sources of structure (G10). *Status: G1, G2, G4, G7, G8 settled; G11–G14 proposed.*

v25 — confirmations, a check of the assembly claim, and bowls keeping labels

- *G11.* Confirmed: the word is multiplicity, used loosely.

- *G13*. The first part is confirmed: links by label criteria are derived at read time. The second part is accepted on this explanation: an immutable older atom cannot be edited to record a later one, so it never stores a forward pointer. “What was A used in?” is answered by searching for atoms whose backward link names A , and that search result is the forward view.
- *G12, refined and checked*. V_{derived} is the sum of an atom’s vectors, with a magnitude and a direction that assist it in higher dimensions, and the stated expectation is that atoms mostly assemble with others of similar magnitude and direction. Checked against the assembly energy $U_{AB} = -\frac{r_{AB}}{2}(3 + \hat{n}_A \cdot \hat{n}_B)/(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}$, $\varepsilon_{AB} = r_A + r_B$:
 - *Direction: supported*. Aligned pairs pull exactly twice as strongly as opposed ones at every separation (ratio 2.000 at $d = 0, 0.5, 1, 3$), and in a random population of 300 atoms the best partner averages 12° to 19° away against about 90° for a random partner.
 - *Magnitude: conditional*. For an atom of $r = 0.60$ the best partner has $r = 0.30$ at contact ($d = 0$), 0.42 at $d = 0.5$, 0.69 at $d = 1$, and 1.00 at $d = 3$. Only at fixed total strength do balanced pairs win, with $|U|$ at $d = 1$ of 0.1768 for $(0.5, 0.5)$, 0.1485 for $(0.7, 0.3)$ and 0.0636 for $(0.9, 0.1)$. In the population the best partner’s strength gap is smaller than a random partner’s at $d = 0$ and $d = 1$ (0.27 and 0.24 against 0.32) and larger at $d = 5$ (0.42 against 0.31).

So direction is a clean grouping rule, while magnitude is a weight whose effect depends on separation relative to the units’ own extents. Two cautions: $|V_{\text{derived}}|$ is uncapped and grows with Q and with multiplicity, so a many-vector atom is heavy in the product and in ε_{AB} ; and the direction is undefined at cancellation, where atom coherence is 0. This is a candidate for R22: read an atom’s single $(r, \hat{n})$ from V_{derived} , which the paper already allows for operators that need one pair.

- *G14, revised*. A bowl may keep labels as well, since labels only identify the nature of atoms and $\uplus$ has no reason to read them; they stay available to anything that wishes to. Consequence: $\uplus$ still discards count in what it reads, and content-level quantities see one seat, but if per-copy labels ride along, count survives in them whenever the labels differ. The pack and unpack map between a bowl and a Collapsed atom (R39) is then lossless exactly when per-copy labels are distinct, for instance identifiers. Bowl and Collapse stay nearly identical but not the same: repeats enter a Collapsed atom’s content, so S , $\mathcal{E}$ and V_{derived} see them, and they do not enter a bowl’s. Proposed default, unconfirmed: labels carried on a seat of equal content merge as a set, equal values coinciding and distinct ones both kept.
- *G5, G6, defaults proposed*. The paper already uses “label” in the sense meant (“direction is a label here,” “a label on the relation”), so the individual meta vector is a label and atoms are grouped by their labels, with no new noun and no “type”; “ordering” replaces “sorting.” If a noun is wanted, “category” has zero uses in the paper, against “class” with four and “kind” with ten. The timestamp is an absolute scalar for age and expiry, with the clock-face vector optional for grouping by time of day.

Status: G11, G13 confirmed; G12 refined; G14 revised; G5, G6 defaults proposed; import awaiting go-ahead.

4 Merged directly, without an ideas entry

These went straight into the paper at your direction and are already in `Pattern_Arithmetic.v15_22.tex`: the Pāṇini expansion (pratyāhāra and vipratīṣedha as structural parallels, with the Ingerman 1967 attribution of “Pāṇini–Backus form” and the oversimplification caveat); the paragraph on summing separate realities; the quantum-mechanics/general-relativity paragraph, deliberately hedged so that it does not claim Pattern Arithmetic bridges them; and the N/D addition (filed afterwards as R5).

5 What remains, in dependency order

1. **Import the meta-vector taxonomy into Paper 1 (R15), after settling what remains in R45** (G3, G5, G6, G9 and G10 remain; G1, G2, G4, G7 and G8 were settled in v24, and G11 to G14 were confirmed or revised in v25). It unblocks R14, which closes Gap 2 and gives location, the timestamp and expiry registers, and V_{derived} a home.
2. **Then merge the v17 resolution (R14)**, replacing the paper’s “Open: location as a sort” paragraph. Decide R19 (expired contributions) at the same time or defer it explicitly. If R38 is adopted, v17’s “per vector” registers should read “per core.”
3. **Decide the scope questions (R27, R28)**: Axiom 7’s metrics and the costumes, and whether to split. These also decide whether R12, R25, and R41–R43 still matter for Paper 1.
4. **The atom’s structure (R20–R22, R36–R40)**. The word is settled (R38), and cores of any size, the empty atom, and Collapse keeping cores distinct are merged (R37, R44; v15_24). What remains is mixed sorts, per-vector τ_i (R36), which closes R20 in the same step. R21 and R22 can each be closed by a sentence, whether a stated refusal or a stated rule, and the commuting test (R39, R40) can then be run.
5. **Package the Inference-bound items (R17, R30–R35)** as one coherent addition to the Inference paper: the candlestick, the technique-selection argument, the spherical Douglas–Peucker adaptation, the sphere-packing distance, and meta-vector grouping.
6. **Small residuals**: the one-sentence R26 clause, the review’s presentational items (R29), shell-indexed energy (R23), the antipodal special case (R24), and whether S_{atom}/N earns a name.